

Md Minhazul Islam

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🌐 Portfolio 🔗 LinkedIn 🐙 GitHub 📊 Kaggle 🆔 ORCID 🎓 Google Scholar

About Me

I have recently completed my BSc in Computer Science and Engineering from Rajshahi University of Engineering and Technology and working as a remote Research Assistant at Qatar University. My research interests center on large language models (LLMs), multimodal analysis, Medical data analysis, agentic AI, retrieval-augmented generation (RAG), natural language processing, and medical text analysis. I am passionate about advancing AI-driven systems and have contributed to several research projects and publications in these areas.

Research Interests

- Natural Language Processing (NLP), Computational Linguistics, Low-Resource Language Processing
- Agentic AI-based system implementation with RAG
- Medical Text Analysis, Healthcare Informatics, Semantic Search
- Large Language Models (LLMs), Multimodal Learning, Contextual Embeddings
- Explainable AI, Model Interpretability, Topic Modeling

Publications

Journal Articles

- [J1] Rashid, M. H. O., Jubaer, M. T., **Islam, M. M.**, et al. (2024). *Live and Mediated User Engagements: A Comparative Dataset from Two Bengali Audio-Story-Based YouTube Channels*. *Data in Brief, Elsevier*.

Conference Papers

- [C1] **Islam, M. M.**, Jubaer, M. T., Srizon, A. Y. (2023). *Navigating Bengali Linguistics: Insights from Machine and Deep Learning Perspectives for Categorization of Sentences*. *26th ICCIT*.
- [C2] **Islam, M. M.**, Jubaer, M. T., Rashid, M. H. O. (2024). *Consistency of Contextual Embedding in Literary Texts*. *ICISSET 2024*.
- [C3] Jubaer, M. T., Chowdhury, B., **Islam, M. M.**, Rashid, M. H. O. (2024). *Detecting Discrepancy to Enhance the Accuracy of Sentiment and Emotion Analysis*. *STI 2024*.
- [C4] Jubaer, M. T., Farzana, M., Chowdhury, B., Zaman, M. S. U., Srizon, A. Y., **Islam, M. M.** (2024). *Analyzing Audience Engagement in Esports: Sentiment and LLM-Based Topic Insights*. *ICCIT 2024*.
- [C5] **Islam, M. M.**, Jubaer, M. T., Srizon, A. Y., Hossain, M. A. (2024). *Explainable AI-Driven Improved Disease Prediction through Symptom Analysis*. *ICCIT 2024*.
- [C6] **Islam, M. M.**, Zedni, A.A, Jubaer, M. T., Srizon, A. Y. *Multilingual Hate Speech and Slang Detection: Evaluating Hybrid Transformers Against Zero-Shot and Few-Shot LLMs Struggling with Short and Low-Context Texts*. *ICCIT 2025*.

Under Review/Submitted

- [J1] **Islam, M. M.**, Jubaer, M. T., Srizon, M. A. Y., Aloteibi, S., Moni, M. A. (2026). *Encoder-Encoder Cross Attention Based Contextual Sarcasm Detection*. *Journal of Big Data*. (Under Review)
- [J2] **Islam, M. M.**, Jubaer, M. T., Khandakar, A., Sarker, S., Rahman, S., Mia, M. M., Ayari, M. A., Noori, H. (2026). *YTClickbait21K: Human-Annotated Multimodal Dataset for YouTube Clickbait Detection Across Diverse Channels and Content Categories*. *Scientific Data (Nature Portfolio)*. (Under Review)
- [J3] Khandakar, A., Michelson, D. G., **Islam, M. M.**, Mia, M. M., Ayari, M. A., Khan, K. (2026). *Unified Privacy Score: A Multi-Dimensional Framework for Attack-Aware Privacy and Utility Evaluation*. *IEEE Transactions on Information Forensics and Security*. (Under Review)
- [J4] **Islam, M. M.**, Mia, M. M., Khandakar, A., Ayari, M. A., Noor, M. T. A., Khan, M. S., Noori, H. (2026). *ICFNet: An Incongruence-Aware Cross-Modal Unified Framework for YouTube Clickbait Detection with Multimodal LLM Comparison*. *Information Processing Management*. (Revision Submitted)

Experience

Research Assistant

Qatar University, Doha, Qatar

March 2026 – Present

- Working on Agentic AI and LLM
- Specialized in Multi-modal analysis

Junior Research Scientist

Global Public Health Research Foundation (GPHRF), Dhaka, Bangladesh

January 2026 – May 2026

- Working on Medical Data analysis
- Implementing AI models in Public Health domain

AI Engineer

makebell Limited, Hong Kong (Remote)

June 2025 – September 2025

- Developed legal-domain RAG + Agentic AI translation system
- Specialized in prompt engineering, LLM integration, scalable backend

Research Assistant

Young Learners Research Lab, RUET (On-site)

Jan 2023 – August 2025

- NLP & deep learning research on Bengali/English datasets
- Co-authored multiple conference/journal publications

Research Assistant

Computational Linguistics Research Lab (Remote)

Jun 2023 – Present

- Word/contextual embeddings, topic modeling, stylometric analysis of literary texts

AI Trainer – Mathematics and Coding

Outlier (Remote)

June 2024 – September 2025

- Dataset creation and quality control for LLM training

ICT Instructor

Talent IT and ICT, Rajshahi, Bangladesh

2020 – 2022

- Taught ICT to secondary and higher-secondary students

Education

BSc Computer Science and Engineering

Rajshahi University of Engineering and Technology

CGPA: 3.42/4.00

Jan 2020 – August 2025

Higher Secondary Certificate

Rajshahi Govt. City College

GPA: 5.00/5.00

May 2017 – Apr 2019

Projects

Contextual Sarcasm Detection (Undergraduate Thesis)

GitHub: [minannu/sarcasm](#)

Python, PyTorch, Transformers

2025

- Developed an encoder-encoder cross-attention model to detect sarcasm in social media text, jointly modeling comments and their context for improved accuracy.
- Implemented custom pre-training strategies to enhance contextual understanding and capture subtle linguistic incongruities.
- Achieved strong performance on benchmark political sarcasm datasets, demonstrating effectiveness of specialized attention mechanisms in context-dependent NLP tasks.

Clickbait Content Detection

GitHub: [minannu/clickbait](#)

Python, Scikit-learn, PyTorch, CLIP, mBERT

2024–current

- Designed a multimodal (text + image) system for detecting clickbait on Youtube data by combining BERT embeddings with CLIP visual features.
- Engineered feature fusion techniques and fine-tuned classifiers, achieving 95% detection accuracy on large-scale datasets.
- Applying Multimodal RAG using VLM to understand the capabilities of VLM's.

Medical Triage Classification

Python, Custom BERT, Regex, Medical Dictionary

2024–2025

- Built a BERT-based hybrid classification model with numeric data to categorize triage notes into critical, urgent, and non-urgent classes using domain-specific pre-training.
- Integrated rule-based methods with medical dictionaries to improve classification consistency and handle rare terminology.
- Achieved 83% accuracy, providing a foundation for AI-assisted clinical decision support systems.

US Election 2024 Data Analysis

GitHub: [kowshik24/us-election-2024](#)

Python, LLM, GraphRAG, t-SNE, HDBSCAN

2025

- Conducted semantic search and topic modeling on Twitter datasets related to the US election to identify trends and public sentiment.
- Leveraged GraphRAG for knowledge retrieval and combined clustering techniques (t-SNE + HDBSCAN) to visualize and interpret user communities.
- Extracted actionable insights into social media influence and misinformation patterns, demonstrating applications of NLP and graph-based AI in political analysis.

Technical Skills

Languages: Python, C, C++, PHP

ML/DL Frameworks: PyTorch, Scikit-learn, Transformers, BERT, RAG, LoRA, Quantized LLM Fine-tuning

Tools: Git, AWS, Prompt Engineering, Pandas, NumPy, Matplotlib

Honors and Awards

- Robi Datathon 3.0: Finalist & Team Leader (May 2024) [\[Link\]](#)
- Dataverse Challenge: Top 15 Finalist (2023) [\[Link\]](#)
- CUET ETE DAY 2023 ML Competition: Top 10 Finalist (2023) [\[Link\]](#)

Leadership & Volunteer Experience

- Officer Cadet, Bangladesh Military Academy (Jan–Apr 2022)
- Social Worker & IT Advisor, BD Clean Bagmara (2024–Present)
- AI Research Mentor, University Research Community (2023–Present)

Certifications

- Green Card: Eligible for Military Officer Course (ISSB, 2021)
- Certificate of Appreciation: 26th ICCIT (IEEE, 2023)
- Paper Presentation: ICISSET 2024, ICCIT 2024

Language Proficiency

Bengali: Native **English:** Fluent

References

1. **Dr. Md. Ali Hossain**, Professor & Head, CSE, RUET
Email: ali.ruet@gmail.com Phone: +880-1959600498  [Website](#)
(Research Supervisor: Medical Text Analysis)
2. **Mohammad Harun Or Rashid**, PhD Student, University of Louisiana at Lafayette
Email: harunmr9@gmail.com Phone: +880-1713205228  [Website](#)
(Supervisor: Text Analytics, Digital Humanities)
3. **Md. Azmain Yakin Srizon**, Assistant Professor, CSE, RUET
Email: azmainsrizon@gmail.com Phone: +880-1790187189  [Website](#)
(Thesis & NLP Supervisor)